

Data Analytics

Research Assignment 1

1. A Database is a repository which stores information

Databases are essential in the modern world for collecting, storing, organising and ordering business information which can assist in making smart decisions that serve customers better, make business operations efficient, reduce expenditure and increase turnover.

Examples where Databases are used;

- a) Financial Services & Credit providers use existing clients' and new applicants' KYC and Credit history to tailor product offerings, market and up-sell add-on products to increase their debtor books.
- b) Social Media companies collect users' site and page visits ~~and~~ as well as engagements to feed the users* with aligned advertisements and recommend similar content consumed.

2. A Data Warehouse is a central system which stores cleaned, organised historical Data, pulled from multiple source systems.

It differs from a Database in that, it houses Databases that are relational.

A company would use a ~~Database~~ Warehouse to easily access structured, precise and concise Data.

3. A Data Lake is a large storage system designed to store large volumes of Data in its original untamed raw form, unfiltered

Businesses might prefer Data Lakes for use in real-time analytics

4. Data Mart is a system which contains a narrow subject of Data and is located within a Data Warehouse.

A Data Mart would be used by a specific department within a business, whose information is niche and specific to that department's unique function.

5. Structured Data is organised information located in a Database, in tables within a schema. It is consistent and has a predefined format. eg. Customer names, contact numbers and ~~also~~ assigned ~~unique~~ customer identifier.

Semi-structured Data is information which can or might not reside in a Database and is not formatted in the same way as information found in Databases. eg. User information found on an XML document, a text file that uses custom tags that describe the structure of the Data it contains. XML (eXtensible Markup Language).

6. Data Modeling is the process of creating a visual representation/diagram of how Data is stored, organised and related in a Database system.

It is important to make sure that Data is logically structured effectively so, before building the actual Database, reducing errors and making it easier for systems to retrieve information.

Examples of Data Models;

a) Conceptual Data Model is a high-level view that shows the type of Data that is needed by the business without getting into ~~the~~ advanced technical details.

b) Physical Data Model is a detailed technical design that shows how ~~the~~ the Database will be built, which will include table names, columns and the types of Data.

7. Data Mining is the process of sorting through large Datasets to discover hidden patterns, correlations and useful insights and trends using machine learning and statistics.

Businesses use it to predict customers' future buying habits, improve marketing techniques and reduce operational risks.

Examples

a) Retail Recommendation Engines: Retail stores analyse past purchase history to suggest items customers might like to buy on their next visit.

b) Credit Card Fraud Detection: Banks monitor spending patterns in real-time to flag unusual transactions that do not match a customer's typical behaviour.

8.) A DBMS is a system ~~that enables~~ software that enables a user to create, read, update and manage Data in Database

DBMS examples:

a) MySQL is an open-source relational Database which can be used to store structured user accounts for a website or a blog

b) PostgreSQL is an advanced open-source relational known for handling complex queries in businesses such as banks whose finance systems require high accuracy and complex Data calculations

c) Oracle is a commercial relational Database used to manage large-scale Data for multinational corporations & government institutions.

d) MongoDB is a document-based Database which ~~does~~ not use SQL; used for storing unstructured Data such as user profiles or real-time chat messages.

9. ETL is a Data pipeline process used to move Data from various sources into a centralized location; i.e. a Data Warehouse

- a) Extract - pulling raw ~~Data from~~ unstructured Data from different sources such as spreadsheets or web servers.
- b) Transform - cleaning, formatting and structuring ~~the~~ Data, to make it consistent and usable, e.g. correcting dates or client names and deleting duplicates.
- c) Load - saving the clean, transformed Data onto a Data Warehouse for analysis.

ETL is important as it ensures that analysts work with clean, accurate and standardized Data rather than messy files in various locations

10. a) CSV (Comma-Separated Values) is a simple text format where Data is stored in rows separated by commas, used when working with simple, small tabular Datasets that need to open easily in spreadsheet software.

b) JSON (JavaScript Object Notation), a text format that uses human-readable key-value pairs to store nested or semi-structured Data, used for exchanging Data between web apps and API (App Programming Interface)

c) Parquet is storage formatted in columns, optimized for heavy Data processing. It is used for storing huge Datasets in big Data platforms where fast Data retrieval and compression is required

d) Avro is a row-based storage format that includes a Data schema inside the file itself, used for streaming real-time Data where schemas might change over time.

11. AI is the simulation of human intelligence in machines programmed to think, learn, and solve problems.

The difference between Narrow and General AI is that, Narrow AI performs a single specific task very well, while General AI is able to understand, learn and apply knowledge across any intellectual task just like a human.

Examples;

- a) Narrow AI; a chess playing computer
- b) General AI; a humanoid robot chef that can cook and write poetry and have a conversation naturally across any topic

12. Machine Learning (ML) is a subset of AI focused on building systems that learn from data to improve systems' performance without being explicitly programmed for every rule.

The difference; In traditional programming, humans write explicit rules and input data to get an output, while in Machine Learning, humans input data and correct outputs, and the computer figures out the rules itself.

- a) Supervised Learning: The model trains on labeled data (input and correct output pairs) to learn how to predict outcomes for new data.
- b) Unsupervised Learning: The model explores unlabeled data to find hidden structures, groupings or patterns on its own.
- c) Reinforcement Learning: A bot learns to make decisions by performing actions in an environment and receives rewards or penalties.

13) Deep Learning is an advanced branch of Machine Learning that uses multi-layered artificial neural networks to analyse complex patterns. Within the broad concept of AI, Machine Learning is a subset of AI and Deep Learning is a specialised subset of Machine Learning.

11 Neural Networks are computing systems inspired by the human brain, made of interconnected nodes, organised into input, hidden and output layers. Data passes through the layers where each connection adjusts its weight until the network correctly identifies or processes the information.

14. NLP is the branch of AI which helps computers understand, interpret and generate human language.

Examples;

- a) Spam filters: automatically sorts unwanted junk emails out of the main inbox.
- b) Predictive text / auto correct: suggests the next word when one types on their cell phone.
- c) Virtual Assistant: voice-activated helpers that understand spoken commands to answer questions or set alarms.

15. A Large Language Model, LLM, is an advanced deep learning model trained on massive amounts of text data, to understand and generate human-like text.

a) The Difference between LLM & ML model is that, a Machine Learning Model might predict a single numerical value, while an LLM generates rich, contextual natural language text sequences.

b) 2 well-known LLMs and their uses:

* Chat GPT: used for drafting text, answering questions and summarising documents.

* Claude: used for creative writing, code assistance and analysing long research documents.

16. Generative AI is AI that can create brand-new content based on the training Data it has learned from. It does this by analysing patterns in massive Datasets and uses probabilistic algorithms to generate new, original content when given a prompt.

a) Midjourney: Generates digital artwork and realistic images from text prompts

b) ElevenLabs: Converts written text into hyper-realistic human voice audio

c) Sora: Generates short, high-definition video clips from text descriptions.

17. A Training Dataset is a collection of Data used to teach an AI model how to recognise patterns and make accurate predictions. The quality of training is important to ensure the model makes reliable, accurate predictions, to produce flawless Data. If the model's ~~output~~ performance is low, the output means that the AI learned prejudices that can lead to unfair discriminatory outcomes in decision making processes.

18. Computer Vision is a field of AI that enables computers to extract meaningful information from digital images and videos by scanning numbers assigned to the broken down images in grids of pixels. These numbers detect edges, shapes, colours and objects

a) Facial Recognition: Unlocking a smartphone using your face.

b) Autonomous Driving: Self-driving cars detecting pedestrians, road signs and other vehicles on the road.

19. An Application Programming Interface is a set of rules and protocols that allow different software applications to communicate with each other

Analogy: A waiter takes an order (1. request), delivers it to the Kitchen (2. system) and brings your food (3. response)

Application Programming Interfaces are important because they allow different software platforms, apps and databases to share information and feature seamlessly without needing to rebuild everything from scratch.

20. An API Key is a unique string of characters used to authenticate a user or program calling an API. Its purpose is to track who is using the API, prevent unauthorised access and manage usage limits.

API Keys must be stored securely in environment variables or secret management systems and never in public code repositories

If API Keys are exposed publicly, unauthorised users can steal access, steal Data and cause business financial losses and cause reputational or legal damage

21. REST API is an architectural style for building web APIs using standard HTTP requests.

REST stands for Representational State Transfer

Main HTTP methods:

- a) GET ; retrieve Data from a server
- b) POST ; sends new Data to a server to create a resource
- c) PUT, updates an existing resource completely on the server
- d) DELETE ; removes a resource from the server

22. JSON (JavaScript Object Notation) is a lightweight, text-based Data Format used for storing and transporting Data.

It is commonly used when working with APIs Data exchange because it is easy for both humans to read and write, and easy for machines to break down and generate across any programming language

JSON object example

"name": "South Africa"

"capital": "Tshwane"

"currency": "Rand"

"region": "Africa"

23. Prompt Engineering is the art/skill/science of structuring text instructions so that an AI tool gives the most accurate and useful response possible. It is an important skill because it bridges the gap between the ambiguous human intent and the precise output of a computer, saving time and improving output quality.

Examples;

a) Coding - Poor prompt: "Fix my code"

- Improved prompt: "Review this Python code snippet for syntax errors and explain how to optimise the loop for better performance"

b) Writing - Poor prompt: "Write about dogs"

- Improved prompt: "Write a 3-paragraph informative guide on the nutritional needs of golden retriever puppies for new pet owners."

24. A **MAP** function is a built-in programming function that takes an array or list, applies a specific operation to every single item in it and returns a new collection with the results.

Example; Python

```
numbers = [1, 2, 3, 4]
```

```
squared = list(map(lambda x: x ** 2, numbers))
```

```
# Result: [1, 4, 9, 16]
```

25. Cloud Computing is the delivery of computing services such as Data storage, servers, Databases and networking) over the internet.

The main Cloud service models;

- a) IaaS (Infrastructure as a Service): Renting raw computing infrastructure like virtual servers and storage, e.g, Amazon Web Services (AWS).
- b) PaaS (Platform as a Service): Providing a managed platform and environment for developers to build and deploy apps without worrying about underlying hardware, e.g, Google App Engine.
- c) SaaS (Software as a Service): Delivery fully functional software applications entirely over the web on a subscription basis, eg, Microsoft 365.

26. A Version Control System is a system that tracks and manages changes to code and documents over time

Git is a popular distributed version control system widely used as it allows teams to collaborate seamlessly, track who made changes and roll back errors safely.

- a) Commit; a snapshot of changes made to files
- Branch; an independent line of development that allows a user to work on features without altering the main code base.

27. Jupyter Notebook is an interactive web-based application that allows users to create and share documents containing live code, equations, visualisations and narrative text. It is commonly used with Python. Jupyter Notebook is popular among Data Analysts and Data Scientists because it allows for code to be run in small testable blocks (cells) and view Data visualisations immediately right next to the notes.

28. Python is a high-level, easy-to-read general-purpose programming language, popular for its clean, simple syntax and massive ecosystem of specialised libraries which make it ideal for handling math, statistics and machine learning.

Python Libraries:

- a) Pandas: used for data manipulation and analysis using Databases
- b) NumPy: used for high-performance mathematics operations on large multi-dimensional arrays.
- c) Matplotlib: used for creating static, animated and interactive Data Visualisations.
- d) Scikit-learn: used for building traditional machine learning models, like regression and classification.

29. Data Privacy is the branch of Data management concerned with how personal information is collected, stored, shared and used.

Data Privacy is important as it protects individuals from identity theft, unauthorised tracking and invasion of personal privacy.

General Data Protection Regulation (GDPR) is a strict data privacy and security law created by the European Union (EU). It gives

EU citizens control over their personal Data and imposes heavy fines on organisations that violate privacy rules, applying to all member countries of the EU and the European Economic Area (EEA), as well as any global company handling EU citizens' data.

30. Data Ethics is a moral framework guiding the responsible collection, analysis, and use of data and AI systems.

Examples of Unethical use of Data and AI;

- a) Data; Selling user browsing data without consent: Violates personal trust and privacy boundaries
- b) AI; Using biased algorithms for hiring decisions unfairly discriminates against certain demographic groups.
- c) AI; Deepfake manipulation for fraud or defamation misleads the public and ruins reputations using fake media.

31. Data Governance is the overall management of the availability, usability, integrity and security of data used in an enterprise.

Key components of a good data governance framework;

- a) Data quality; ensuring data is accurate and reliable
- b) Data security; protecting sensitive data from breaches
- c) Data stewardship; assigning clear roles and accountability for data management
- d) Compliance; adhering to legal regulations and company policies.

32. Bias in AI is systematic errors in an AI model that create unfair, prejudiced outcomes for certain groups of people

Example: Racial bias in healthcare.

- A widely used healthcare risk-prediction algorithm in the U.S.A favored white patients over black patients because it used healthcare spending as a proxy for medical need. Since less money is historically spent on black patients, the model underestimated their care requirements, impacting over 200 million citizens.

33. Key differences between a Data Analyst, Data Scientist, Data Engineer and an AI Engineer.

ROLE	PRIMARY ROLE	Key SKILLS	Common Tools
Data Analyst	Interprets past Data to help businesses make everyday decisions	SQL, basic stats, Data visualisation	Excel, Tableau, Power BI, Python
Data Scientist	Builds predictive models and uses advanced math to find deep insights	Machine learning, advanced statistics, Python/R	Python, R, Jupyter, SQL
Data Engineer	Builds and maintains systems for collecting and storing Data	Database architecture, ETL, pipelines, cloud computing	SQL, Spark, Airflow, Snowflake
AI Engineer	Builds and deploys AI models and software applications	Software engineering, deep learning, API integration	Python, PyTorch, TensorFlow, LangChain

34. Business Intelligence (BI) is the technology, tools and practices used to collect, integrate, analyse and present business Data in visual dashboards and reports

Two popular BI tools:

- Microsoft Power BI and
- Tableau, turn complex rows of Data into clear, interactive charts and dashboards, allowing executives to spot trends quickly and make informed, fact-based business decisions

35. ~~D~~ General Data Protection Regulation and the countries to which it applies.
It's surprising to learn that it is not applied globally, but comforting to know that there is an effort to align South Africa's GDPR and POPIA obligations through shared governance, risk management and data protection practices.
36. Version Control System. I used w3schools.com to read over the description of a Git too, in simple english.
37. Data and AI, if ~~used~~ applied incorrectly will take away from human beings creativity as efficiency and quick reward is favoured by lazy people and profit.
Used correctly to assist in both business and everyday life, AI can assist with monotonous tasks.

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